Advanced Lecture on Astronomy 1
宇宙物理学特論1

Toshikazu Onishi
大西利和

Course Objectives/ Overview

- This course provides an overview of the evolution and cycle of interstellar matter and the related observation methods. The history and present state of the observation technology of millimeter and submillimeter astronomy will be introduced, and the current understanding of star and planet formation based on the observations will be lectured. The latest results that are being clarified by ALMA will also be presented.

- To understand the evolution of the interstellar matter
- To understand the mechanism of electromagnetic radiation from the interstellar matter
- To understand the basic principles of star formation
- To understand the basics of radio astronomy.
- Current and unresolved problems in star/planet formation are summarized, and methods for solving these problems are discussed.

Methods

- Lectures on Radio Astronomy focused on interstellar medium and star formation

- Student’s presentations
  - This will be used for the Grade Evaluation.

What are the topics for the presentation?

- Describe an unresolved astrophysical question you are interested in.

  or any question related to astrophysics in some sense

  - What is the issue? What is the background of the question?
  - Why is it intriguing? Why is it important?
  - Why does it remain unresolved?

  e.g., mm-wave "detection" of WIMP dark matter

- Then discuss how existing and/or anticipated radio (to far-infrared) observing facilities can address the question.
  - You can think of *any* telescopes like ALMA, JVLA, MeerKAT, ASKAP, CHIME, SKA, Nobeyama, LMT, ASTE, JCMT, GBT, SPT, ACT, FAST, GBT, VERA, VLBA, EHT, CCAT-p, LST/AtLAST, AAT, Planck, LiteBIRD, Herschel, etc. etc..
  - Use of the existing (or growing) archival data is also fine.
  - Describe your method and analysis plan specifically
    - What frequency band(s), spectral line(s), angular resolution, spectral resolution, required sensitivity, cadence, polarization, etc.
    - What are key physical quantities you want to obtain from the analysis? How do you use these key physical quantities to answer the question you specified?

Then, how can I make my presentation?

- You will have 15 minutes for your presentation, which includes 5 minutes for discussions.
- Prepare for your presentation in e.g., pptx, keynote, pdf.
  - Please include relevant figures, references, etc.

Lecture schedule

Subject to change

Lec #	Date	Topics
1	Apr. 14 (2026)	Introduction of this lecture
2	Apr. 21	Introduction: ISM and star formation
3	Apr. 28	Tracing atomic/molecular hydrogen
4	May 12	(continue)
5	May 19	Ionized region/ Dust observations
6	May 26	Overall Picture of Star Formation: Observations and Basic Concepts
7	Jun. 2	Low-mass star formation
8	Jun. 9	Star and Planet Formation Revealed by Radio Astronomy
9	Jun. 16	(continue)
10	Jun. 23	Buffer
11	Jun. 30	Students' presentations (6 presentations)
12	Jul. 7	Students' presentations (6 presentations)
13	Jul. 14	Students' presentations (6 presentations)
14	Jul. 21	Students' presentations (6 presentations)
15	Jul. 28	if required

Time-line for your presentation

Declaration before June 22: You may submit your preferred date among dedicated presentation days (June 30, July 7, July 14, and July 21).

No declaration before June 22: The presentation date will be assigned to either June 30, July 7, July 14, and July 21. If you have a preference for the date, please submit your request by June 22.

Submission of your presentation

Submit your presentation file by the Friday of the week prior to the lecture in which you are presenting. Minor revisions are allowed at the time of presentation. Submitted files will be placed on the Moodle for everyone to read, so please take the time to read other people‘s presentation files and ask questions/comments when presentation. I accept either English or Japanese for the presentation. Please prepare at least your presentation slides in English.